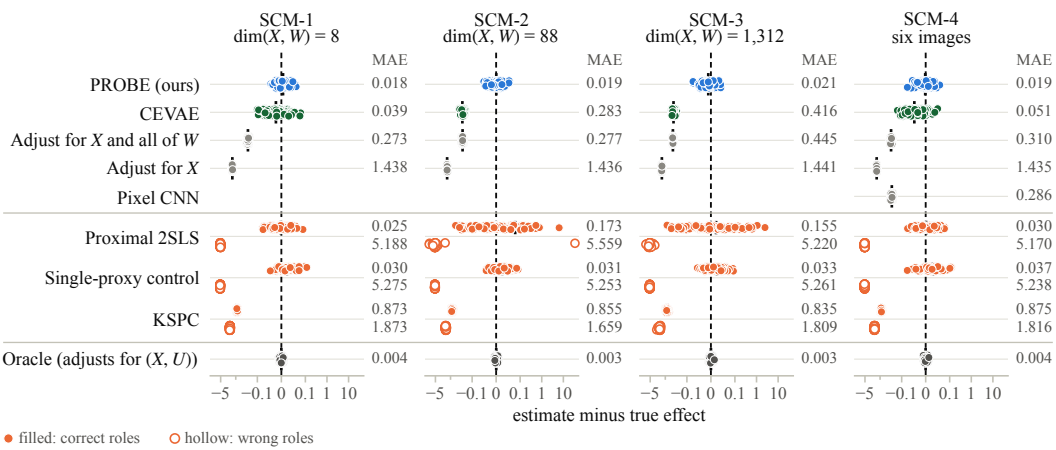

SCM-1
dim(X, W) = 8
SCM-2
dim(X, W) = 88
SCM-3
dim(X, W) = 1,312
SCM-4
six images
MAE
PROBE (ours)
0.018
0.019
0.021
0.019
CEVAE
0.039
0.283
0.416
0.051
Adjust for X and all of W
0.273
0.277
0.445
0.310
Adjust for X
1.438
1.436
1.441
1.435
Pixel CNN
0.286
Proximal 2SLS
0.025
5.188
0.173
5.559
0.155
5.220
0.030
5.170
Single-proxy control
0.030
5.275
0.031
5.253
0.033
5.261
0.037
5.238
KSPC
0.873
1.873
0.855
1.659
0.835
1.809
0.875
1.816
Oracle (adjusts for (X, U))
0.004
0.003
0.003
0.004
−5 −0.1 0 0.1 1 10
estimate minus true effect
filled: correct roles
hollow: wrong roles